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TCE–603

B. TECH. IN CIVIL ENGINEERING

(SIXTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

WATER RESOURCES ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Classify the various types of dams. Distinguish clearly between rigid and non-rigid dams. (CO1)

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OR

(b) Design an elementary profile of a gravity dam from following data : (CO1)

R.L. of the base of the dam = 400 m

R.L. of water stored up to top of dam = 440 m

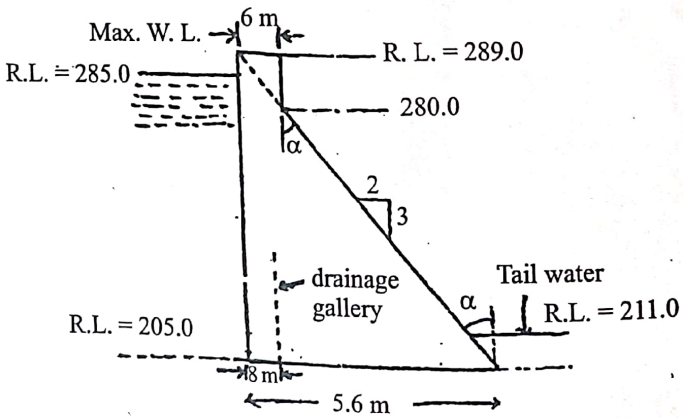
Uplift factor = 0.75

Specific gravity of concrete = 2.4

2. (a) What are the factors on which selection of site for a dam depends ? (CO1)

OR

(b) The figure shows the section of a gravity dam (non-overflow portion) built of concrete. Calculate the maximum vertical stress at the heel and toe of the dam. Assume weight of concrete as 23.5 kN/m^3. Neglect earthquake effects. (CO1)



3. (a) Discuss in brief various modes of failure of a gravity dam. (CO1)

OR

(b) Distinguish clearly between a low gravity dam and high gravity dam. Determine the critical height of a low gravity dam of concrete, taking the specific gravity of concrete as 2.40 and allowable compressive stress as 3340 kN/m^3. (CO1)

4. (a) Enumerate the different types of earthen dams and draw neat sketches showing each type. (CO2)

OR

(b) For a homogeneous earth dam 52 m high, and 2 m free board, a flow net was constructed and following results were obtained :

Number of potential drops = 25

Number of flow channels = 4

The dam has a horizontal filter of 40 m length at its downstream end. Calculate the discharge per metre length of the dam if the coefficient of permeability of the dam is 3×10^{-3} cm/sec. (CO2)

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5. (a) Discuss in brief the causes of failure of earthen dam. (CO2)

OR

(b) In order to find the factor of safety of the d/s slope during steady seepage, the section of a homogeneous earth dam was drawn to a scale of 1 cm = 10 m and the following result were obtained on a trial circle :

Area of N-rectangle	= 12.15 sq.cm
Area of T-rectangle	= 6.50 sq. cm
Area of U-rectangle	= 4.02 sq. cm
Length of arc	= 11.60 cm

The dam material has the following properties :

Effective angle of internal friction	= 26°
Unit cohesion	= 20 kN/m^2
Unit weight of soil	= 20 kN/m^3

Determine the factor of safety of the slope. (CO2)

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